

Development and evaluation of a novel two-stage exponential current ramping strategy for low-temperature self-cold start of PEM fuel cells

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ABSTRACT

Proton exchange membrane fuel cells (PEMFCs) are promising clean energy technologies for mobile and stationary applications. However, achieving rapid and reliable self-cold start at subzero temperatures remains a critical bottleneck for PEMFC commercialization due to ice formation and irreversible performance degradation. This study addresses a key gap in cold-start research: the lack of systematic analysis on how the shape of current ramping profiles affects the coupled thermal, electrochemical, and phase-change dynamics. To address this knowledge gap, a transient three-dimensional multi-physics model is developed that incorporates electrochemical reactions, mass and heat transfer, and water-ice phase change. Three ramping strategies—linear, exponential, and a novel two-stage exponential—are evaluated under equivalent charge input to explore their impact on voltage stability, thermal response, and ice formation dynamics. The simulation results reveal that exponential ramping causes a rapid voltage drop of 0.2 V within 15 s due to premature ice accumulation, while the two-stage exponential strategy delays voltage decay by approximately 20 s and maintains a final voltage of 0.73 V, 4.3 % higher than that of the linear strategy. Additionally, the two-stage profile reduces the final average ice volume fraction in the cathode catalyst layer by approximately 9.3 %, improves temperature uniformity, and suppresses vertical ice propagation into gas diffusion layer. These improvements stem from the sequential co-ordination of heat generation and water management across both stages of ramping. This work provides theoretical guidance for optimizing current ramping strategies and offers insights into the design of intelligent control algorithms for PEMFC cold start under extreme conditions.

1. Introduction

Amid escalating global environmental challenges such as climate change, ozone depletion, acid rain, and air pollution, the global energy landscape is undergoing an unprecedented transformation. The intensified development and utilization of clean energy, along with the pursuit of energy decarbonization, has become a dominant trend in global energy policy and technology [1]. Proton exchange membrane fuel cells (PEMFCs) have gained significant attention as clean and efficient energy conversion devices for automotive, stationary, and portable power applications [2]. Compared with battery electric vehicles (BEVs), fuel cell electric vehicles (FCEVs) offer faster refueling, longer range, and better cold-climate adaptability—making them attractive for large-scale decarbonized mobility, particularly in regions with renewable hydrogen supply chains [3]. These advantages make them ideal candidates for next-generation energy systems. However, the commercialization of PEMFCs, especially in vehicular applications, is critically constrained by

their poor performance under sub-zero temperature conditions, commonly referred to as cold start [4].

During cold start, the water produced by the electrochemical reaction at the cathode can freeze before being removed from the catalyst layer (CL) or gas diffusion layer (GDL). The formation of ice leads to several detrimental effects, including blockage of gas pathways [5], increased mass transport resistance [6], reduced electrochemically active area [7] and mechanical stress within the membrane electrode assembly (MEA) [8]. These phenomena may ultimately lead to irreversible damage to cell components. Such issues highlight that the low-temperature environment poses a serious challenge to the normal operation of PEMFC. Therefore, achieving a successful and efficient cold start is essential for ensuring the durability and reliability of PEMFC systems in real-world, all-weather environments.

To address the challenges posed by sub-zero environments, the development of effective cold start strategies has become a focal point in PEMFC research [9]. A well-designed cold start strategy can not only significantly enhance the cold start performance of PEMFCs, but also

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Nomenclature		Subscripts and superscripts	
A	area (m^2)	a	anode
a	water activity	c	cathode
C	molar concentration (mol m^{-3})	desub	desublimation rate
C_p	specific heat capacity ($\text{J mol}^{-1} \text{K}^{-1}$)	eff	effective
D	mass diffusivity ($\text{m}^2 \text{s}^{-1}$)	ele	electrical
F	Faraday's number 96487 (C mol^{-1})	eq	equilibrium state
i	current density (A m^{-2})	freeze	freeze
k	thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$)	g	gas phase
I	average current density (A m^{-2})	H_2	hydrogen
$\dot{m}$	mass flow rate (kg s^{-1})	H_2O	water vapor
P	pressure (Pa)	i	gas species
Q	total charge transfer (C)	ice	ice phase
R	universal gas constant 8.314 ($\text{J mol}^{-1} \text{K}^{-1}$)	ion	ionic
S	source term of governing equations	lq	liquid water
s	volume fraction	O_2	oxygen
T	temperature (K)	p	pressure
t	time (s)	ref	reference state
Y	mass fraction	sat	saturation state
		sub	sublimation rate
Greek letters		Abbreviations	
η	overpotential (V)	BP	bipolar plate
γ	water phase change rate (s^{-1})	CCL	cathode catalyst layer
ε	porosity	CL	catalyst layer
κ	Intrinsic permeability (m^2)	CGDL	cathode gas diffusion layer
θ	contact angle ($^\circ$)	CH	channel
λ	membrane water content	GC	gas channel
σ	conductivity (S m^{-1})	GDL	gas diffusion layer
ρ	density (kg m^{-3})	MEA	membrane electrode assembly
μ	dynamic viscosity (N s m^{-2})	PEM	proton exchange membrane
ξ	stoichiometric ratio	PEMFC	proton exchange membrane fuel cell
ϕ	potential (V)	UDF	user-defined function

improve their durability and service life [10]. At present, several strategies have been proposed to improve cold start performance, including external pre-heating [11], chemical heaters [12], resistance heating [13], purge cycles, and current-loading methods [14]. In practical applications, Naganuma et al. [15] proposed a rapid preheating method that limits the gas supply to increase concentration overpotential and reduce efficiency. This method was incorporated into the 2008 Toyota FCHV-adv, enabling successful and repeated cold starts at -30°C . However, there remains significant room for improvement. Maruo et al. [16] developed a 3D fine-mesh cathode flow field combined with a purging technique to enhance water management during cold start. Although this design has been successfully applied in the Mirai-FCV, further improvement is required to meet commercialization standards. The second-generation Toyota Mirai (2020–) has been commercialized and retains cold start capability at -30°C . However, instead of relying solely on self-heating, it integrates a grid-type external heater to assist ice melting and ensure hydrogen supply to the stack [17]. To provide sufficient heat to the stack, Li et al. [18] employed external heating wires placed under the cathode rib for localized heating. This approach enabled a successful cold start with minimal external energy input, though the limited heating power led to voltage instability. Zhou et al. [19] used an electric heater to supply additional thermal energy, achieving successful start-up at -20°C . However, the low heating efficiency caused heat losses during the process, limiting its broader applicability. Luo et al. [20] employed a coolant preheating system to warm the stack before cold start at -30°C . While this method ensured cold start success, the high thermal mass of the coolant required prolonged heating time, resulting in increased energy consumption and longer startup delays. Zhan et al. [21] proposed a staged preheating

strategy, conducting a series of experiments using air preheating, coolant preheating, and endplate heating. They found that while air preheating was slow, heating the endplates proved efficient and time-saving. Sun et al. [22] considered a catalytic hydrogen/oxygen reaction method to supply heat. In this approach, hydrogen and oxygen are introduced into a shared reaction chamber, where they react directly to form water on the catalyst layer, releasing heat without generating current—essentially acting as a short-circuit heater. However, the additional hardware introduces extra weight, volume, and complexity in installation and control, limiting its practical deployment.

Although the auxiliary heating strategies mentioned above can partially meet cold start requirements at low temperatures, the use of external heating devices increases system complexity and cost, reduces operational flexibility and robustness, and may negatively impact cell longevity and reliability [23]. In contrast, self-heating strategies based on internal heat generation from the stack reactions have attracted increasing attention due to their independence from external thermal sources [24]. Compared with structural modification or complex auxiliary heating systems, strategy-level optimization offers a more flexible, cost-effective, and system-compatible solution. The electrochemical self-heating through current loading is the most energy-efficient and system-integrated approach, as it utilizes the internal ohmic and reaction heat generated by the fuel cell itself. Stepwise current loading, in particular, is a widely studied and practically applicable method due to its simplicity and controllability [25]. In this approach, the applied current is increased in discrete steps to gradually raise the internal temperature while minimizing ice formation and thermal stress.

However, most existing research focuses on the selection of parameters such as current step magnitude, duration, and current thresholds

[26]. The loading curve of the current ramp remains an underexplored yet potentially impactful factor. Different ramping profiles can lead to significantly different heat generation rates, local temperature distributions, and phase-change dynamics [27]. A rapid ramp may quickly warm up the system but also risks localized overheating and performance non-uniformity. Conversely, a gradual or feedback-controlled ramp may enhance thermal stability and reduce ice accumulation but at the cost of longer startup time. Meanwhile, in order to ensure the reliability and fairness of the research results, it is necessary to guarantee that the total input energy of PEMFC is the same.

To address this knowledge gap, this study aims to systematically investigate the influence of current ramping profile shape on the low-temperature PEMFC cold start behavior under equal charge constraints. A three-dimensional, transient multi-physics model is developed, incorporating electrochemical reactions, heat transfer, phase change, and mass transport. Three current ramping strategies—linear, exponential, and a newly proposed two-stage exponential—are evaluated and compared in terms of voltage evolution, temperature evolution, and ice formation. The findings presented here offer new insights into the dynamic electrochemical–thermal interactions during cold start.

2. Model and methodology

To analyze the self-cold start behavior of PEM fuel cell under sub-zero conditions, a transient three-dimensional multi-physics model was established. The model incorporates key physical processes including electrochemical reactions, heat and mass transfer, and water/ice phase change. This section outlines the geometric configuration, meshing, governing equations, boundary and initial conditions, and the overall simulation procedure.

2.1. Geometric model and mesh

Fig. 1 illustrates the complete computational domain of the PEMFC and details of the mesh structure. The corresponding geometrical parameters are listed in Table 1. A three-dimensional model of a PEMFC with four parallel flow channels was constructed to simulate its transient cold start behavior. The geometry includes all major functional components of a typical PEMFC: anode and cathode flow fields, GDLs (0.2 mm), CLs (10 μm), PEM, (50 μm , corresponds to Nafion 212, a standard PFSA membrane widely used in PEMFC modeling and experiments), and bipolar plates (BPs, 2 mm). The total active area was set to 500 mm^2 . A parallel flow field layout was adopted to avoid interference caused by

Table 1
Geometrical parameters.

Symbol	Parameter	Value	Unit
L_x	Bipolar plate width	10	mm
L_y	Bipolar plate length	50	mm
L_z	Bipolar plate height	2	mm
l_1	Edge width	1.5	mm
l_2	Channel height	1	mm
l_3	Channel width	1	mm
l_4	Ridge width	1	mm
n_a/n_c	Channel number: Anode/Cathode	4	—
h_{GDL}	GDL thickness	0.2	mm
h_{CL}	CL thickness	0.01	mm
h_{PEM}	PEM thickness (Nafion 212)	0.05	mm
S	MEA area	500	mm^2

the crossflow of reactant gases in opposite directions.

To ensure the accuracy and reliability of the simulation results, a grid independence study was conducted by evaluating the effect of mesh resolution on the electrochemical performance of the PEMFC model under a fixed output voltage of 0.70 V. MEA is the most core component of hydrogen fuel cells and the core site for multiphase transport and electrochemical reactions. Therefore, five different grid configurations were calculated by changing the number of grid layers of MEA, with the total number of elements ranging from 233,510 (Grid 1) to 712,140 (Grid 5). The corresponding current densities at the specified voltage are shown in Fig. 2.

As the mesh density increased, the calculated current density exhibited a noticeable decline and then gradually stabilized. Specifically, the current density dropped significantly from 51.1 mA cm^{-2} (Grid 1) to 39.8 mA cm^{-2} (Grid 2), and further to 35.1 mA cm^{-2} (Grid 3), after which the change between Grid 4 and Grid 5 became negligible (<1%). This indicates that numerical diffusion and discretization error were significantly reduced beyond Grid 3. Based on these results, Grid 3 with 574,150 elements was selected for all subsequent simulations. It offers a good compromise between computational efficiency and numerical accuracy.

2.2. Conservation equations

Unlike the mass transfer characteristic that liquid water can be freely discharged under normal temperature conditions, the formation of ice during low-temperature cold start-up will block the gas channels and occupy the electrochemical reaction points, resulting in a significant increase in gas transmission resistance. Against this background, the

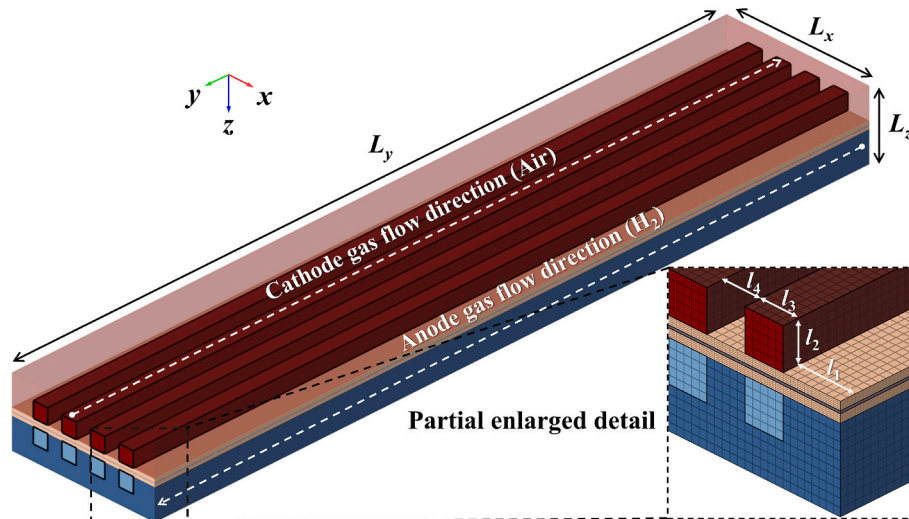


Fig. 1. PEMFC computational domain and mesh detail.

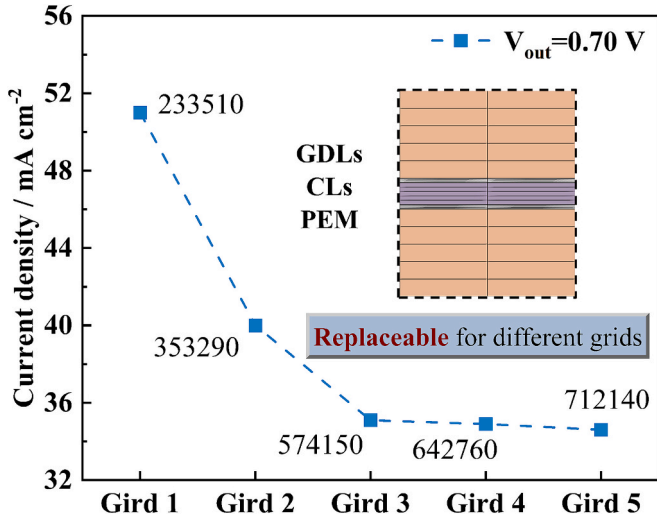


Fig. 2. Grid independence validation.

coupling of physical processes at different scales becomes particularly complex. To accurately simulate the transient cold start behavior of PEM fuel cells, a series of reasonable physical assumptions were adopted to simplify the mathematical model while preserving the essential physical mechanisms [28]. These assumptions are as follows:

1. All gases are treated as incompressible and ideal.
2. The flow is assumed to be laminar ($Re < 2300$).
3. Hydrogen and oxygen are impermeable through the PEM.
4. The GDLs and CLs are considered isotropic.
5. The densities of liquid water and ice are constant.
6. Phase change from ice directly to water vapor is not considered [29].
7. External heat sources are excluded from the cold start process.

Based on these assumptions, the model solves a set of coupled transient conservation equations, including mass, momentum, charge, energy, species transport, and phase change. The equations are formulated in a three-dimensional domain representing the PEMFC's internal structure and are solved using a finite-volume method.

Mass equation: Solve in the flow channel, CL, and GDL.

$$\frac{\partial}{\partial t} [\varepsilon(1 - s_{lq} - s_{ice})\rho_g] + \nabla \cdot (\rho_g \vec{u}_g) = S_m \quad (1)$$

where ε is the porosity, s_{lq} and s_{ice} are the volume fractions of liquid water and ice, respectively, ρ_g and u_g denote the density and velocity of the gas mixture, respectively.

Momentum conservation: Solve in the flow channel, CL, and GDL.

$$\frac{\partial}{\partial t} \left[\frac{\rho_g \vec{u}_g}{\varepsilon(1 - s_{lq} - s_{ice})} \right] + \nabla \cdot \left[\frac{\rho_g \vec{u}_g \vec{u}_g}{\varepsilon^2(1 - s_{lq} - s_{ice})^2} \right] + \nabla p_g = \mu_g \nabla \cdot \left[\nabla \left(\frac{\vec{u}_g}{\varepsilon(1 - s_{lq} - s_{ice})} \right) \right] + \nabla \cdot \left[\left(\frac{\vec{u}_g^T}{\varepsilon(1 - s_{lq} - s_{ice})} \right) \right] - \frac{2}{3} \mu_g \nabla \cdot \left(\frac{\vec{u}_g}{\varepsilon(1 - s_{lq} - s_{ice})} \right) + S_u \quad (2)$$

where p_g is the gas-phase pressure, and μ_g is the dynamic viscosity of the ideal gas mixture.

Species transport: Solve in the flow channel, CL, and GDL.

$$\frac{\partial}{\partial t} [\varepsilon(1 - s_{lq} - s_{ice})\rho_g Y_i] + \nabla \cdot (\rho_g \vec{u}_g Y_i) = \nabla \cdot (\rho_g D_i^{eff} \nabla Y_i) + S_i \quad (3)$$

where Y_i and D_i are the mass fraction and mass diffusivity of species i

(H_2 , O_2 , H_2O). The effective diffusivity D_i^{eff} is corrected for porosity and tortuosity of the GDL/CL structure [30].

Liquid water: Solve in CL and GDL.

$$\frac{\partial(\varepsilon s_{lq} \rho_{lq})}{\partial t} + \nabla \cdot (\rho_{lq} \vec{u}_{lq}) = \nabla \cdot (\rho_{lq} D_{lq} \nabla s_{lq}) + S_{lq} \quad (4)$$

where ρ_{lq} and D_{lq} denote the density and diffusivity of liquid water, respectively.

Phase change: The ice-phase model is applied to all porous electrodes, but not considered in the gas channels. Only mass transfer between the gas phase and the ice phase is taken into account.

$$\frac{\partial(\varepsilon s_{ice} \rho_{ice})}{\partial t} = S_{ice} \quad (5)$$

where ρ_{ice} is the density of ice, and the mass source term of the ice phase is calculated as follows:

$$S_{ice} = \begin{cases} 0, & C_{H_2O} < C_{H_2O}^{sat} \\ \gamma_{desub}(C_{H_2O}^{sat} - C_{H_2O}), & C_{H_2O} > C_{H_2O}^{sat} \text{ and } T < T_{freeze} \\ \gamma_{sub}(C_{H_2O}^{sat} - C_{H_2O}), & s_{ice} > 0 \text{ and } T = T_{freeze} \\ 0, & s_{ice} = 0 \text{ and } T > T_{freeze} \end{cases} \quad (6)$$

where C_{H_2O} represents the local water vapor concentration, $C_{H_2O}^{sat}$ is the local water saturation concentration, T_{freeze} is the freezing temperature, γ_{desub} and γ_{sub} denote the water desublimation and sublimation rate parameters, respectively. Since the sublimation process (i.e., the direct phase transition from ice to water vapor) is not considered in this model, γ_{sub} is set to zero.

Charge conservation: the core of electrochemistry is to calculate the rate of anode and cathode reaction. The driving force behind these reactions is the surface overpotential. Therefore, two potential equations are solved.

$$\nabla \cdot (\sigma_{ele} \nabla \phi_{ele}) + S_{ele} = 0 \quad (7)$$

$$\nabla \cdot (\sigma_{ion} \nabla \phi_{ion}) + S_{ion} = 0 \quad (8)$$

where σ is the electrical conductivity and ϕ is the electric potential. The electron potential is solved in the CL, GDL, and BP domains, while the proton potential is solved in the CL and PEM domains.

Electrochemical Reaction Kinetics: The local current density in the catalyst layer is governed by the Butler-Volmer equation.

$$i = i_0 [\exp(\frac{\alpha_a F \eta}{RT}) - \exp(\frac{-\alpha_c F \eta}{RT})] \quad (9)$$

where i_0 is the exchange current density, α_a , α_c are the anodic and cathodic transfer coefficients, η is the overpotential, F , R , and T have their usual meanings.

These equations are solved in a fully transient manner to reflect the time-dependent evolution of temperature, species distribution, and ice formation under varying current loading conditions.

2.3. Boundary and initial conditions

To accurately replicate the low-temperature self-cold start process of the PEMFC, appropriate boundary and initial conditions are applied

across the computational domain, covering thermal, mass transport, and electrochemical aspects.

Fully dried hydrogen and air are supplied to the anode and cathode gas channels, respectively. The mass flow inlets are defined with a fixed flow rate for both gases, with uniform velocity and species distribution across each channel. The outlet boundaries are set as pressure outlets with ambient pressure (1 atm), allowing for unidirectional gas and water transport. At the inlet of the flow channels, the mass flow rate can be defined as follows:

$$\dot{m}_a = \frac{\rho_g^a \xi_a I_{ref} A}{2FC_{H_2}} \quad (10)$$

$$\dot{m}_c = \frac{\rho_g^c \xi_c I_{ref} A}{4FC_{O_2}} \quad (11)$$

Where A is the active area of the PEMFC (500 mm² in this study), C_{H_2} , ρ_g^a and C_{O_2} , ρ_g^c are the concentrations and densities of hydrogen and oxygen, respectively, F is the Faraday constant, I_{ref} is the reference current density, ξ_a and ξ_c are the stoichiometric ratios of reactants at the anode and cathode, respectively, defined as the ratio of supplied reactants to the theoretically required amount. In this study, both ξ_a and ξ_c are set to 2.0 for numerical calculations. The anode current collector is grounded ($\phi_{ele} = 0$ V), while a time-dependent current boundary condition is applied to the cathode side, according to the specified ramping profile (linear, exponential, or two-stage exponential). No current flows through the side walls or membrane surfaces. The entire computational domain is initialized at -30 °C to replicate realistic cold start conditions. Before the cold start, a long purge process was assumed to have taken place and the water content in the CL and PEM was initialized to a hydration level corresponding to $\lambda = 6.2$ [29]. The initial current and ice volume fraction are zero before loading. The cell is thermally isolated from external heat sources to simulate self-cold start conditions. Ice

Table 2
Material properties and operating conditions.

Parameter	Value	Unit
Reference exchange current density (353.15 K) [29]: Anode/Cathode	$10^9/10^4$	A m ⁻³
Inlet gas temperature: Anode/Cathode	243.15/243.15	K
Initial ice volume fraction [29]	0	
Initial membrane water content [29]	6.2	
Operating pressure: Anode/Cathode	1/1	atm
Inlet gas relative humidity: Anode/Cathode	0/0	%
Stoichiometric flow ratio [29]: Anode/Cathode	2.0/2.0	
Reference current density: Anode/Cathode	6000/20	A m ⁻²
Contact angle [31]: GDLs/CLs	110/95	°
Intrinsic permeability: GDLs/CLs/PEM [28]	$1.0 \times 10^{-12}/1.0 \times 10^{-13}/1.0 \times 10^{-18}$	m ²
Porosity: GDLs/CLs	0.8/0.53	
Exchange coefficient: Anode/Cathode	0.5/1.0	
Concentration exponent: Anode/Cathode	1.0/1.0	
Reference diffusivity of H ₂ /O ₂ /H ₂ O/N ₂ [29]	$8.67 \times 10^{-5}/1.53 \times 10^{-5}/1.79 \times 10^{-5}/3 \times 10^{-5}$	m ² s ⁻¹
Equivalent weight of PEM [27]	1100	kg kmol ⁻¹
Density of BPs/GDLs/CLs/PEM [28]	2719/500/1000/1980	kg m ⁻³
Electrical conductivity [27]: BPs/GDLs/CLs	$1 \times 10^7/300/300$	S m ⁻¹
Proton conductivity of PEM [28]	1.0×10^{-16}	S m ⁻¹
Thermal conductivity [31]: BPs/GDLs/CLs/PEM	20/1.0/1.0/0.95	W m ⁻¹ K ⁻¹
Specific heat capacity: BPs/GDLs/CLs/PEM [28]	871/900/1100/2000	J kg ⁻¹ K ⁻¹
Liquid water density	982	kg m ⁻³
Ice density [29]	920	kg m ⁻³
CL surface-to-volume ratio	2.0×10^5	m ⁻¹

formation and low-voltage operation both contribute to long-term MEA degradation. In this study, the success or failure of the self-cold start process is evaluated based on two criteria. Output voltage must remain above 0.4 V throughout the ramping period, ensuring electrochemical functionality is maintained; Ice volume fraction in the CCL must stay below 0.3, indicating that severe pore blockage is avoided. While this definition focuses on modeling feasibility under extreme conditions, future extensions may incorporate system-level metrics such as rated power attainment. Table 2 summarizes the material properties and operating conditions.

2.4. Numerical simulation procedure

The numerical simulations were performed using ANSYS Fluent 2024R2. The physical model and governing equations were implemented using the PEMFC module, with additional physical effects (ice formation, current ramping control) incorporated via user-defined functions (UDFs).

The workflow was as follows: Geometry was built in SolidWorks 2022 and meshed in HyperMesh 2022, with mesh quality checked via skewness and orthogonality metrics. A transient solver was used with second-order implicit time discretization and pressure-based coupling. Solution convergence was monitored for residuals and stabilization of key physical variables. Post-processing was carried out in CFD-Post 2024R2, where spatial distributions of temperature, current density, and ice formation were visualized and analyzed.

3. Current ramping profiles

The variation of applied current plays a crucial role in determining the thermal response, ice dynamics, and electrochemical stability of PEMFCs during cold start. In this study, three distinct current ramping profiles are implemented to evaluate their effects on cold start performance: linear, exponential, and a newly proposed two-stage exponential.

3.1. Mathematical definitions

Linear ramping: The linear profile increases the applied current in uniform increments over a fixed duration. This method is commonly used due to its simplicity and ease of implementation. The current as a function of time is defined as:

$$I(t) = I_0 + \alpha \cdot t, 0 \leq t \leq t_{max} \quad (12)$$

where I_0 is the initial current (typically 0 A), α is the ramp rate (A s⁻¹), t_{max} is the total ramping duration. Although straightforward, the linear ramp may result in insufficient early heating or excessive local heating in the later stages, depending on environmental and material conditions.

Exponential ramping: The exponential profile allows for a slow initial rise in current followed by rapid escalation, accelerating the generation of heat as current increases. The current is given by:

$$I(t) = I_{\infty} \cdot (1 - e^{-\beta t}), 0 \leq t \leq t_{max} \quad (13)$$

where I_{∞} is the target maximum current, β is the exponential growth rate constant (s⁻¹). This strategy mimics thermally driven self-acceleration but may risk non-uniform heating or hot spots if the rate is too aggressive, especially in poorly hydrated or frozen zones.

Two-stage exponential ramping: To improve thermal management across the cold start period, a two-segment exponential strategy is introduced. In the first stage, the applied current increases slowly from zero, following an exponential trend characterized by an accelerating rate of change. As the cold start process progresses, the second stage begins at a predefined transition time and continues with a decelerating growth pattern that gradually levels off. The two segments are smoothly connected at a transition time t_s , with both function value and slope

being continuous. The profile is defined as:

$$I(t) = \begin{cases} A \cdot (e^{k_1 t} - 1), & 0 \leq t \leq t_s \\ I_s + C \cdot (1 - e^{-k_2(t-t_s)}), & t_s \leq t \leq t_{\max} \end{cases} \quad (14)$$

where A is the amplitude of the first stage, k_1 , k_2 are the exponential growth rates for the two segments, $I_s = A(e^{k_1 t_s} - 1)$ is the current at the transition point, $C = Ak_1 e^{k_1 t_s} / k_2$ ensures slope continuity. The final current profile combines the benefits of rapid thermal buildup in the initial stage and controlled temperature rise in the later stage, making it well-suited for balancing heat generation and ice suppression during cold start.

All three ramping strategies are implemented in the model via UDFs and are applied to the cathode current collector boundary.

3.2. Current equivalence constraint

To enable a fair and physically meaningful comparison among different current ramping strategies, this study adopts a current equivalence constraint approach based on equal total charge transfer. Instead of applying the same ramping duration ($t_{\max}$) across all strategies, each profile is allowed to evolve over a custom duration such that the integrated current, or total applied charge, remains constant. Notably, no artificial constraint is imposed on the maximum current value instead, each profile is allowed to naturally evolve to its peak current based on its shape and time span. This approach ensures that all ramping strategies contribute the same amount of electrochemical energy during the cold start process, thereby aligning the thermal input and allowing differences in performance to be attributed to the temporal shape of the current profile rather than total energy. Additionally, it also avoids excessive constraints on the optimization space and achieves more realistic physical comparisons.

The linear ramping strategy is used as the baseline to define the reference total charge. With parameters: $I_0 = 0$ A, $\alpha = 0.0025$ A s⁻¹, $t_{\max} = 80$ s, the total transferred charge is calculated as:

$$Q_{\text{total}} = \int_0^{80} I(t) dt = \int_0^{80} 0.0025t dt = 8C \quad (15)$$

This total charge value is then imposed as a common constraint on the other two current strategies. For each case, the upper time limit $t_{\max}$ is iteratively determined such that:

$$\int_0^{t_{\max}} I(t) dt = 8C \quad (16)$$

This ensures that all three profiles deliver equivalent thermal and electrochemical energy over their respective durations. Such an approach not only improves the fairness of comparison but also represents a more realistic control strategy in practical fuel cell systems, where energy budget is a key constraint. The definitions and parameters of the three current ramping strategies are summarized in Table 3 and their profiles are visualized in Fig. 3.

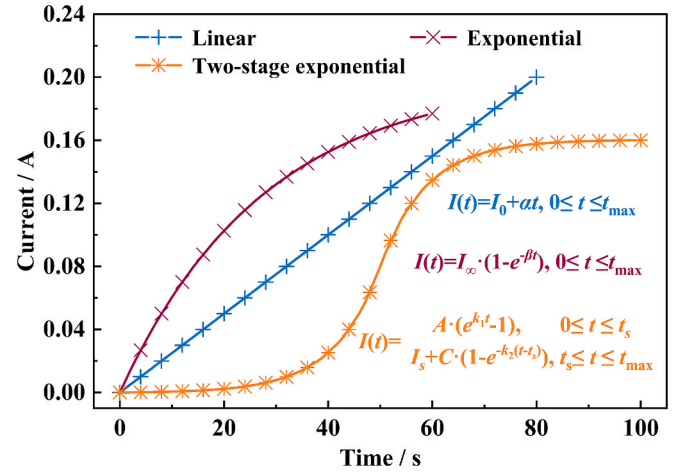


Fig. 3. Visualization of current ramp strategies.

4. Results and discussions

This section systematically evaluates the cold start performance of PEM fuel cells under three distinct current ramping strategies: linear, exponential, and two-stage exponential. Based on the established three-dimensional multi-physics model, the influence of current profiles on key cold start characteristics—such as voltage evolution, temperature evolution, and ice formation—is quantitatively assessed and compared. Through numerical analysis, the benefits and trade-offs of each strategy are examined to provide theoretical support for optimal current design in self-starting scenarios under subzero conditions.

4.1. Model validation

To ensure the accuracy and reliability of the developed numerical model for simulating PEMFC cold start behavior, a validation study was conducted by comparing the simulated cell voltage with experimental results in Ref. [32] under two representative operating conditions: low current density (40 mA cm⁻²) and high current density (80 mA cm⁻²). The operating conditions are consistent with the experimental setup, with an ambient temperature of -30 °C. The initial water content in the CLs and PEM is set to 6.2, and the initial ice volume fraction is zero.

Fig. 4 illustrates the temporal evolution of cell voltage obtained from both simulation and experiment. The output voltage of fuel cells shows an overall decreasing trend with the increase of current. Quantitatively, the average relative error between simulation and experimental voltage data across both conditions remains below 5 %, demonstrating the predictive accuracy of the model. Therefore, this model can serve as a reliable tool for further analyzing the impact of current ramp strategies on the cold start characteristics of PEMFC.

Table 3
Summary of current ramping strategies.

Strategy No.	Description	Mathematical Expression	Time range (s)	Max current (A)	Key parameters
1	Linear ramping	$I(t) = I_0 + \alpha t, 0 \leq t \leq t_{\max}$	0–80	0.2	$I_0 = 0$ A $\alpha = 0.0025$ A s ⁻¹
2	Exponential ramping	$I(t) = I_{\infty} \cdot (1 - e^{-\beta t}), 0 \leq t \leq t_{\max}$	0–60	≈0.18	$I_{\infty} = 0.2$ A $\beta = 0.036$ s ⁻¹
3	Two-stage exponential ramping	$I(t) = \begin{cases} A \cdot (e^{k_1 t} - 1), & 0 \leq t \leq t_s \\ I_s + C \cdot (1 - e^{-k_2(t-t_s)}), & t_s \leq t \leq t_{\max} \end{cases}$	0–100	≈0.16	$A = 0.000259$ A $k_1 = k_2 = 0.1147$ s ⁻¹ $I_s = 0.08$ A $C = 0.0803$ A $t_s = 50$ s

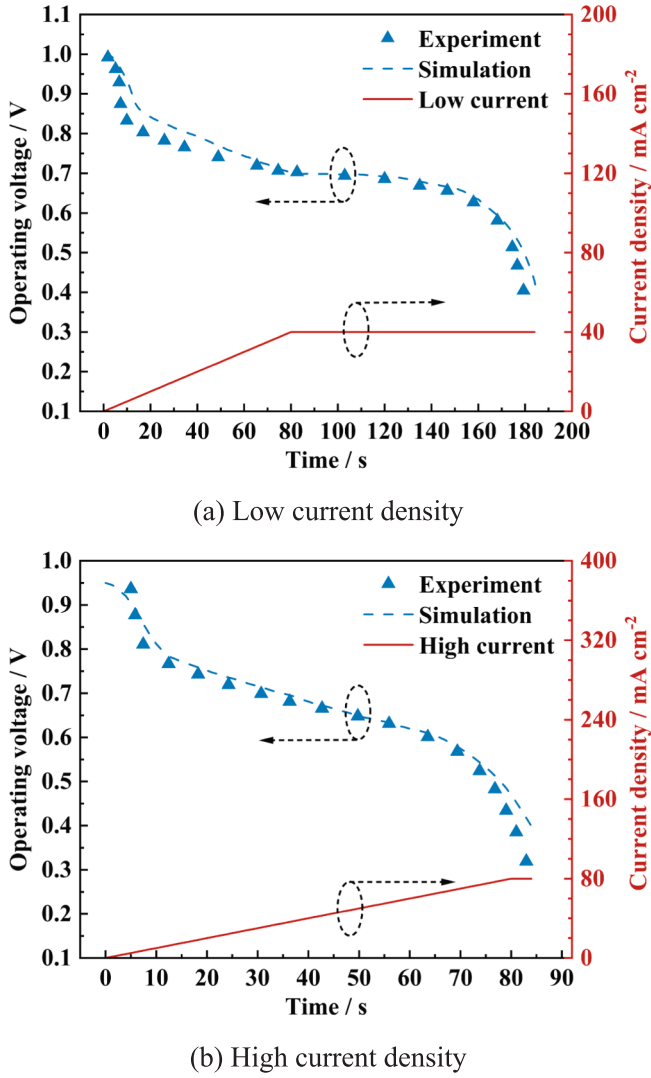


Fig. 4. Numerical model validation.

4.2. Voltage and temperature evolution

To understand how different current ramping strategies influence the electrochemical and thermal stability during cold start, this section examines the coupled evolution of voltage and internal temperature over time. Fig. 5 illustrates the temporal evolution of operating voltage and the temperature at the CCL-PEM interface under different cold start strategies. Although all three current ramping strategies exhibit similar initial trends, their subsequent voltage and temperature profiles reveal fundamentally different internal dynamics driven by heat generation, water production, and phase change phenomena.

In the initial stage ($t < 5$ s), all strategies generate a brief voltage peak close to 1.0 V, corresponding to the dry-start condition with low membrane hydration and minimal ice formation. Simultaneously, a sharp but short-lived temperature spike is observed near the catalyst layer-membrane interface, which results from sudden ohmic and reaction heat release as the current begins to rise. This early thermal impulse is insufficient to raise the overall cell temperature above freezing but plays an important role in preventing immediate ice nucleation. Following this transient peak, the thermal and electrical responses begin to diverge depending on the current ramping profile. Under Strategy 1 (linear ramping), the voltage drops rapidly to 0.85 V and then gradually decreases, stabilizing around 0.70 V by 80 s. The linear increase in current allows for gradual water production and membrane hydration,

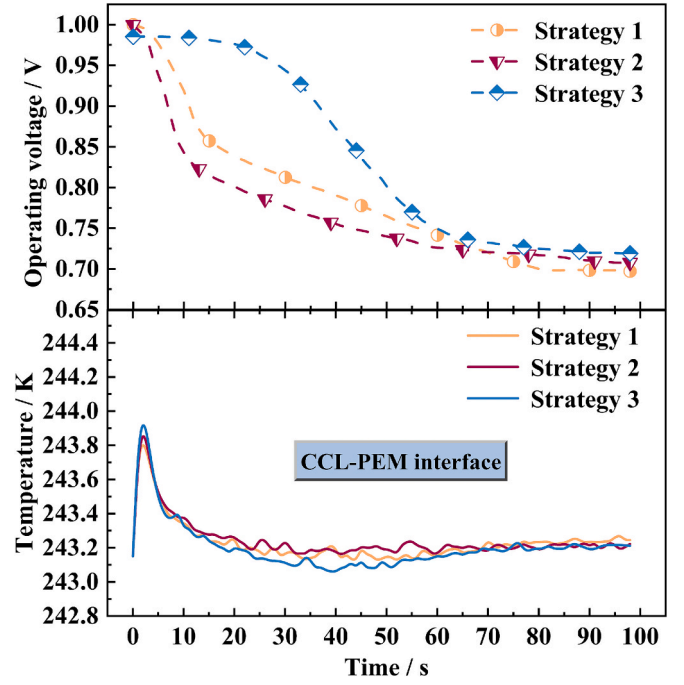
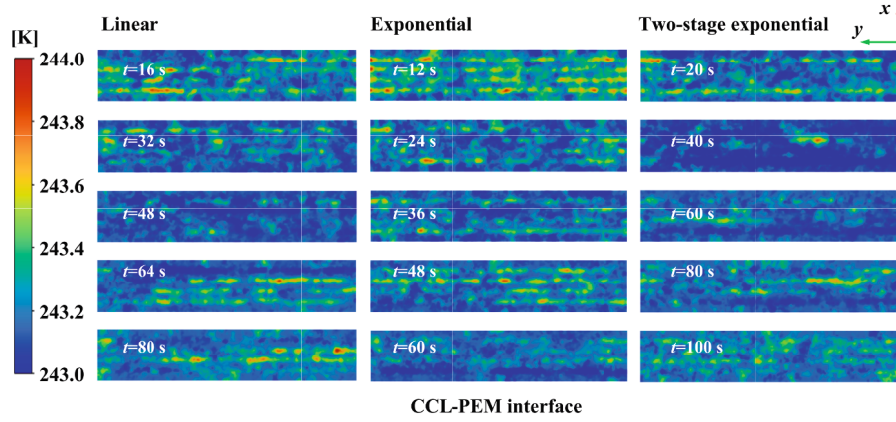


Fig. 5. Changes in operating voltage and temperature during cold start of PEMFC.

which enhances proton conductivity and helps maintain electrochemical performance [33]. However, the slow thermal input prolongs the subfreezing condition of internal components, potentially leading to long-term voltage degradation and an increased risk of cold start failure. Therefore, although the voltage drop in Strategy 1 is slower than that in Strategy 2, it results in the lowest terminal voltage among the three. Strategy 2 (exponential ramping) shows the most dramatic voltage drop in the early stage, with a decline of approximately 0.2 V within the first 15 s. This behavior is attributed to the rapid current increase during the first 20 s, which leads to rapid water accumulation near the CCL. At this stage, localized water freezing occurs, blocking oxygen transport pathways and deactivating reaction sites. Simultaneously, ice formation reduces ionic conductivity, increases ohmic resistance, and intensifies ohmic heating, causing the CCL-PEM interface temperature under Strategy 2 to rise significantly [34]. Strategy 3 (two-stage exponential ramping) exhibits a more favorable voltage profile throughout the cold start process. The initial voltage drop is delayed by approximately 20 s and proceeds at a slower rate, ultimately stabilizing at around 0.73 V, the highest among all strategies. This improvement can be attributed to its segmented current ramping design: the slow current rise in the first stage mitigates rapid water generation and allows time for membrane preheating, thereby reducing the risk of ice blockage during the early cold start. The accelerated second stage then enhances heat generation after a thermal buffer is established, leading to a gradual temperature increase.

Thermal management plays a critical role during the cold start process, as the internal heat distribution directly influences water phase transitions and electrochemical stability. Fig. 6(a) illustrates the spatiotemporal evolution of temperature distribution at the CCL-PEM interface under different current ramping strategies during their respective effective loading durations. In the initial stage, Strategy 2 exhibits a relatively uniform in-plane temperature distribution. This can be attributed to the rapid increase in current, which leads to a more immediate and evenly distributed release of ohmic and electrochemical reaction heat throughout the catalyst layer. This accelerated heating effect mitigates localized cold spots and supports synchronized temperature rise at early times. However, as the current continues to rise



(a) Temperature

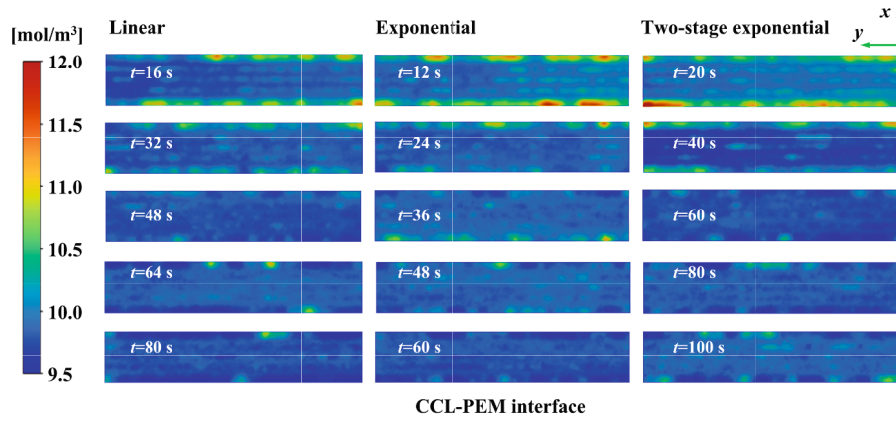
(b) O_2 molar concentration

Fig. 6. Temperature and O_2 molar concentration distribution evolution of CCL-PEM interface under different current ramping strategies during their respective effective loading durations.

exponentially, Strategy 2 subsequently shows a noticeable decline in surface temperature and increasing thermal non-uniformity. This behavior is likely due to premature water production and partial ice formation, which inhibits heat conduction and gas diffusion in certain regions, particularly under the rib. At $t = 40$ s, it is also observed that Strategy 3 exhibits the most pronounced temperature drop among all strategies. This phenomenon is closely related to the very low initial current ramp rate employed in Strategy 3, which leads to insufficient heat generation in the early stage. This observation aligns well with the average temperature curve shown in Fig. 5. Subsequently, the temperature on the surface begins to rise gradually, and thermal uniformity improves progressively. Notably, a clear trend emerges in the mid-to-late phase where the temperature beneath the flow channels is consistently higher than that under the ribs. This is attributed to the preferential reactant gas supply and stronger electrochemical activity in the channel-adjacent regions, which enhances localized heating. In contrast, the rib-covered areas are less reactive, resulting in a persistent temperature differential.

Furthermore, it is important to note that during the cold start period, especially in the middle and later stages, discontinuous temperature distributions occurred on the CCL-PEM plane. The temperature variation is less than 1°C , making it challenging to attribute such a distribution to any specific phenomenon. This unevenness may be related to local water freezing or reactant transport. When product water accumulates and freezes within confined porous regions, it hinders gas transport and the progress of electrochemical reactions, leading to sudden changes in local

electrical conductivity and thermal diffusion, manifested as discontinuous temperature gradients. To elucidate the origin of the discontinuous temperature distribution observed in Fig. 6(a), the corresponding oxygen molar concentration contours at the CCL-PEM interface are provided in Fig. 6(b). While no strict one-to-one correspondence is observed, regions with relatively higher oxygen concentration during the early and middle stages tend to coincide with areas exhibiting elevated temperature. This suggests that reactant availability may play a role in shaping localized heat generation. In later stages, the oxygen distribution becomes more uniform and reduced, possibly due to ice-induced transport limitations, which could be associated with the diminished thermal gradients. Overall, these patterns indicate that reactant transport dynamics might contribute to the non-uniform thermal response during cold start, in addition to water phase change effects.

4.3. Ice formation behavior

Ice formation within the porous components of the MEA is one of the most critical factors influencing the success of PEMFC cold start. The accumulation of ice not only obstructs gas and water transport pathways but also leads to increased ohmic resistance and localized performance degradation. To evaluate how ramping affects freezing dynamics spatially and temporally, this section analyzes the evolution of the ice volume fraction under different current ramping strategies is analyzed to understand how current-driven thermal and mass transport processes affect ice formation and distribution.

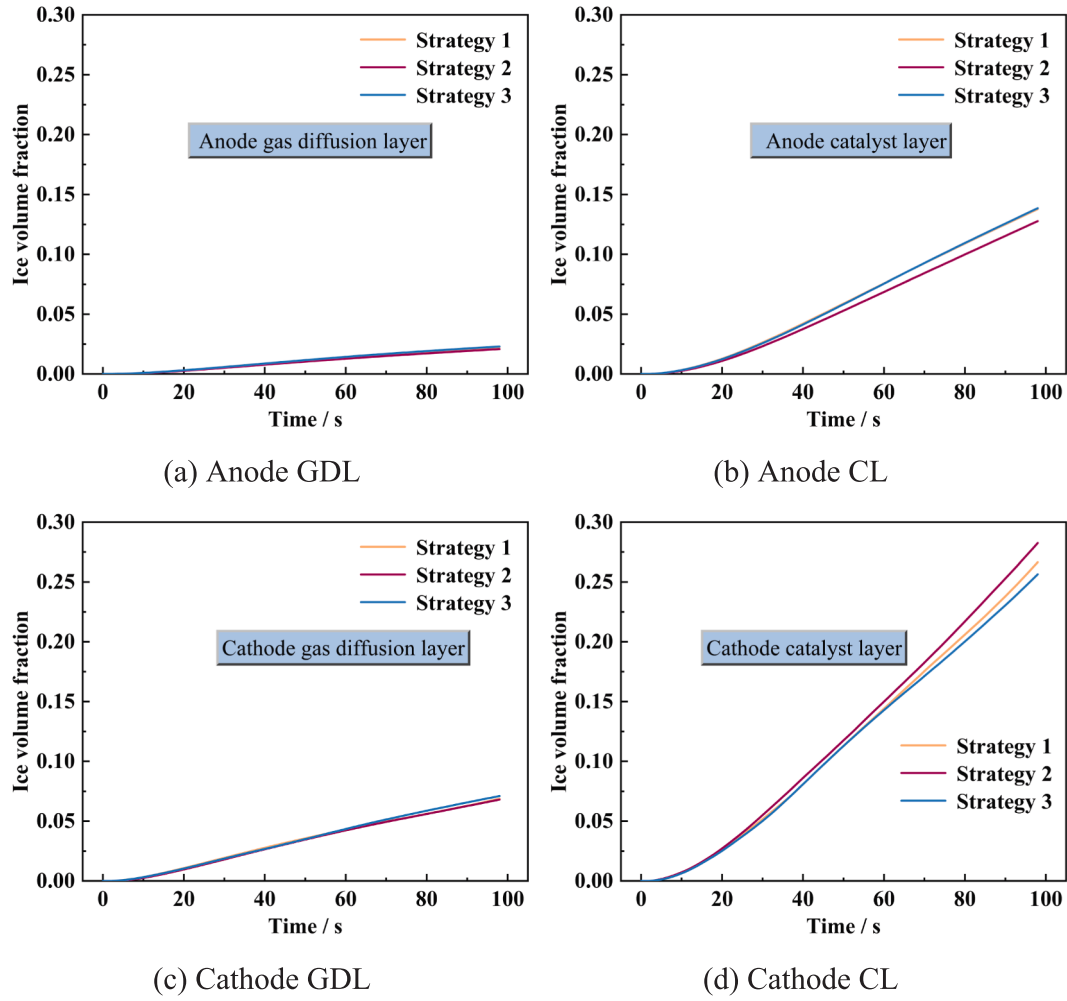


Fig. 7. Evolution of average ice volume fraction in GDLs and CLs under different current ramp strategies.

Fig. 7 presents the evolution of average ice volume fraction in the GDLs and CLs on both anode and cathode sides for the three current ramping strategies. Two clear conclusions can be drawn: first, the amount of ice formed on the cathode side is greater than that on the anode side; second, the catalyst layer accumulates significantly more ice than the gas diffusion layer. As shown in Fig. 7(d), the CCL is the region most prone to icing. This is because product water is primarily generated at the cathode, with a generation rate several times higher than that at the anode. In addition, electroosmotic drag further promotes water accumulation in the cathode, leading to overall greater ice formation on the cathode side [35]. It is also worth noting that, due to the initially preset water content ($\lambda = 6.2$) in both CLs and the membrane, early-stage freezing may occur in the ACL, even though the product water originates from the cathode. Furthermore, the vapor-phase diffusivity of water decreases exponentially with temperature. At $-30\text{ }^{\circ}\text{C}$, the diffusion capacity of water vapor in the CL pores drops to just 5 % of its value at normal temperature [36]. This severely hinders vapor transport from the CL to the GDL, resulting in a very low fraction of ice formed via vapor-phase diffusion. Meanwhile, the saturated vapor pressure of liquid water at $-30\text{ }^{\circ}\text{C}$ is only 1.2 % of that at normal temperature, making evaporation nearly impossible. As a result, liquid water tends to remain trapped within the CL. Taken together, these factors explain why the CL experiences significantly more ice accumulation than the GDL.

Among the three strategies, Strategy 2 (exponential ramping) produces the highest ice accumulation in the CCL, with the average ice volume fraction exceeding 0.28 at the end of the cold start. This can be attributed to the rapid increase in current in the early phase, which leads

to accelerated water generation without sufficient heat generation to prevent freezing. These frozen regions introduce additional mass transfer resistance and partially deactivate reaction sites, which is reflected in the rapid voltage decline observed in Fig. 5. The temperature non-uniformity discussed in Fig. 6 further exacerbates this issue by promoting local supersaturation and ice nucleation [37]. In comparison, Strategy 1 (linear ramping) results in slightly lower ice accumulation in the CCL. The slower increase in current reduces the instantaneous water production rate, prolongs the hydration process, and thus alleviates freezing. However, the overall temperature rise remains limited (Fig. 5), which prevents complete suppression of freezing in low-temperature regions. Strategy 3 (two-stage exponential ramping) demonstrates the most favorable ice mitigation performance in the CCL, with the lowest final ice volume fraction among the three. The low current ramp rate in the initial stage minimizes water production during periods with less heat generation, while the second-stage current boost enhances heat generation when the system has already reached a more favorable thermal condition.

To further elucidate the vertical progression of ice accumulation during cold start, Fig. 8 depicts the local distribution of ice volume fraction along the x - z plane (cross-section at the center of the cell) within the CCL and CGDL for all three current ramping strategies. In the case of Strategy 2 (exponential ramping), ice formation initiates as early as $t = 12\text{ s}$ and rapidly propagates from the catalyst layer interface into the GDL by $t = 24\text{--}36\text{ s}$. This steep progression is driven by the early surge in water production and low temperature, which causes water to freeze not only in the CCL but also gradually invading the CGDL. By $t = 60\text{ s}$, a

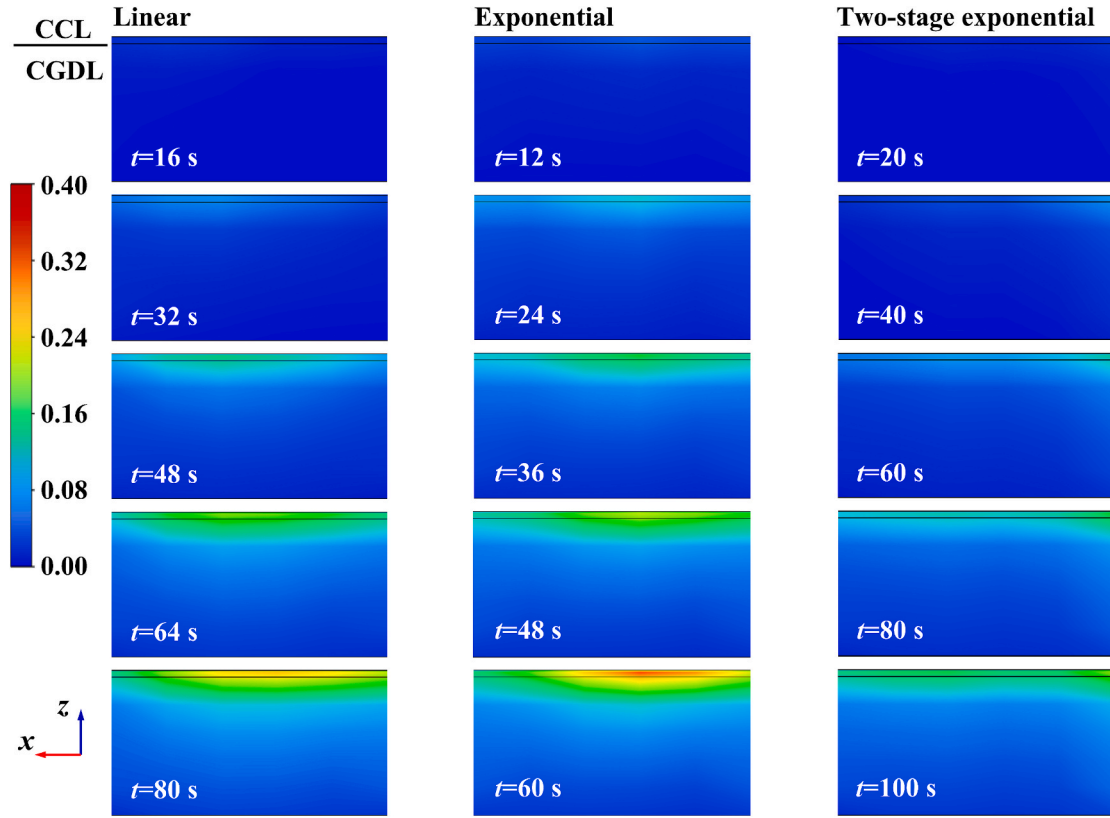


Fig. 8. Local distribution of ice within cathode CL and GDL (x - z section at the center of the cell, scale 0.4 in the x direction).

pronounced ice front is clearly observed, spanning a significant portion of the GDL thickness and exhibiting high ice volume fractions near the CCL-CGDL interface. In contrast, Strategy 3 (two-stage exponential ramping) shows the most favorable spatial behavior. Ice is virtually absent in the CGDL until $t = 40$ s, and even at $t = 60$ – 100 s, there was less ice clearly penetrating into CGDL. The vertical ice distribution analysis highlights that the extent and direction of ice propagation are directly influenced by the current density increase rate. Aggressive early loading leads to ice intrusion into the porous structure, while staged or delayed strategies mitigate this effect.

These results collectively demonstrate that ice formation is strongly dependent on the temporal coupling between water generation and

internal heat generation. Aggressive current ramping before sufficient heating, as in Strategy 2, leads to early freezing and performance deterioration. Conversely, Strategy 3 effectively alleviates the freezing situation by regulating the current in a staged manner, highlighting the critical importance of dynamic strategy design for self-cold start optimization.

Fig. 9 provides the spatial distribution of ice volume fraction across the CCL-PEM interface for each strategy at representative time points. For Strategy 1 (linear ramping) and Strategy 2 (exponential ramping), ice formation begins as early as the initial stage of the cold start and intensifies rapidly during the mid-to-late phase. Initially, ice appears only in localized regions, mainly concentrated near the sides and

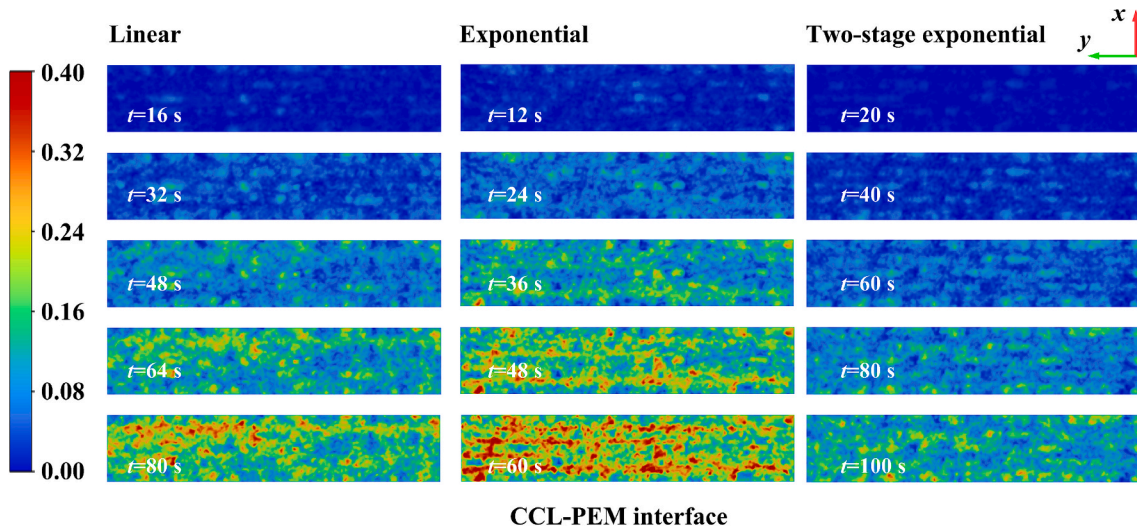


Fig. 9. Ice distribution evolution of CCL-PEM interface under different current ramping strategies during their respective effective loading durations.

beneath the ribs, while water at the CCL-PEM interface remains in a supercooled liquid state. However, as the cold start progresses, ice quickly spreads across the entire plane, forming extensive frozen regions with pronounced high ice volume fraction patches. This is particularly evident in Strategy 2, where by $t = 60$ s, a large continuous area of ice forms, and these ice regions begin to interconnect, significantly reducing the effective contact area between the CCL and the PEM. Strategy 3 (two-stage exponential ramping) shows a distinctly different ice formation trajectory. No obvious ice accumulation is observed until after 60 s, which is much later than the other two strategies. Even at $t = 80$ s, the amount of ice remains noticeably lower than that in the other cases. Although high ice volume fraction regions begin to appear at $t = 100$ s, their overall spatial coverage remains limited. Importantly, Strategy 3 avoids the rapid formation of high-volume fraction ice zones, which are evident in Strategy 2 and partially in Strategy 1. This ensures sustained gas diffusion pathways and minimizes the risk of cold start failure due to ice blockage. Similar ice accumulation behavior has also been reported in experimental studies. Li et al. [38] visualized the cold-start icing process and observed that ice preferentially formed at the interface between the CCL and the microporous layer. This localized ice led to increased interfacial resistance and voltage decay—consistent with our simulation results. The spatial growth of ice observed in Fig. 9 aligns well with their reported patterns of early and concentrated freezing near the reaction zone. It is noted that although the average temperature profiles appear similar across the three strategies (Fig. 5), the resulting ice formation behavior differs markedly. This is attributed to the nonlinear coupling between temperature and water generation: the local timing of water appearance relative to the temperature profile determines whether phase change occurs. Slight shifts in reaction timing can cause water to appear before or after local regions fall below freezing, thus amplifying differences in ice nucleation even under

comparable average thermal conditions.

4.4. Coupled multi-physics mechanism

This section synthesizes the above results to reveal field-coupled interactions. Fig. 10 illustrates the coupled multi-physics mechanism. The cold start process of PEMFCs is governed by the highly coupled interaction of electrochemical reactions, thermal transport, and water phase transitions, where heat affects electricity, ice affects mass transfer and electricity feeds back heat. The dynamic evolution of voltage, temperature, and ice volume fraction under different current ramping strategies collectively illustrates how imbalance among these physical fields can lead to performance degradation or startup failure. This section aims to synthesize those results and understand the distinct behaviors observed among the three current profiles.

The exponential ramping strategy (Strategy 2) highlights the risk of temporal mismatch between heat generation and water accumulation. The rapid current rise in the early stage leads to a sharp increase in product water. However, due to the low initial heat generation and the extremely low temperature, the cathode CL freezes early. Once ice forms, it disrupts gas transport and reaction site availability, thereby accelerating voltage drop. In contrast, the delayed ice formation and relatively uniform voltage drop of the linear ramp strategy (Strategy 1) indicate that this strategy avoids severe mismatches. The two-stage exponential strategy (Strategy 3) demonstrates coordination among thermal, electrochemical, and mass transport processes. In the first stage, a low ramp rate prevents premature water flooding. In the second stage, the system utilizes the heat generated in the early stage to adapt to higher currents without causing excessive icing. Moreover, the temperature and ice distribution observed under exponential and linear strategies further emphasizes the importance of field uniformity. Uneven thermal distribution promotes the formation of local ice, while sudden phase transitions introduce steep resistance gradients—these feedbacks can reduce the stability of the system.

In summary, the success of cold start strategies is governed not only by the magnitude of heat input or current density, but more importantly by the timing and coordination of multiple physical fields. A well-designed ramping strategy should ensure: sufficient preheating before aggressive current loading, controlled water production aligned with thermal capacity, and uniform field distribution across both time and space. These insights provide a theoretical basis for the strategy optimization discussed in this study, and offer design guidance for future self-cold start control algorithms under subfreezing conditions.

5. Conclusions

This study developed a transient, three-dimensional multi-physics model to investigate the self-cold start behavior of PEMFCs under sub-zero conditions. Through comparative analysis of linear, exponential, and two-stage exponential profiles, several key insights have been obtained regarding voltage evolution, heat and mass transport, and ice formation dynamics. The major conclusions are as follows:

- (1) **Voltage response is sensitive to the ramping shape.** Exponential ramping causes a sharp voltage drop of 0.2 V within the first 15 s, due to premature ice formation. In contrast, the proposed two-stage exponential ramping delays voltage decay by approximately 20 s and maintains a final operating voltage of 0.73 V, which is 4.3 % higher than the linear ramping strategy.
- (2) **Temperature uniformity is critical for freezing mitigation.** During cold start, non-uniform heat distribution—particularly under the ribs and in CLs—can lead to uneven water phase transitions and early ice nucleation. The two-stage exponential strategy ensures a more gradual and spatially balanced thermal response. This thermal uniformity limits supersaturation zones,

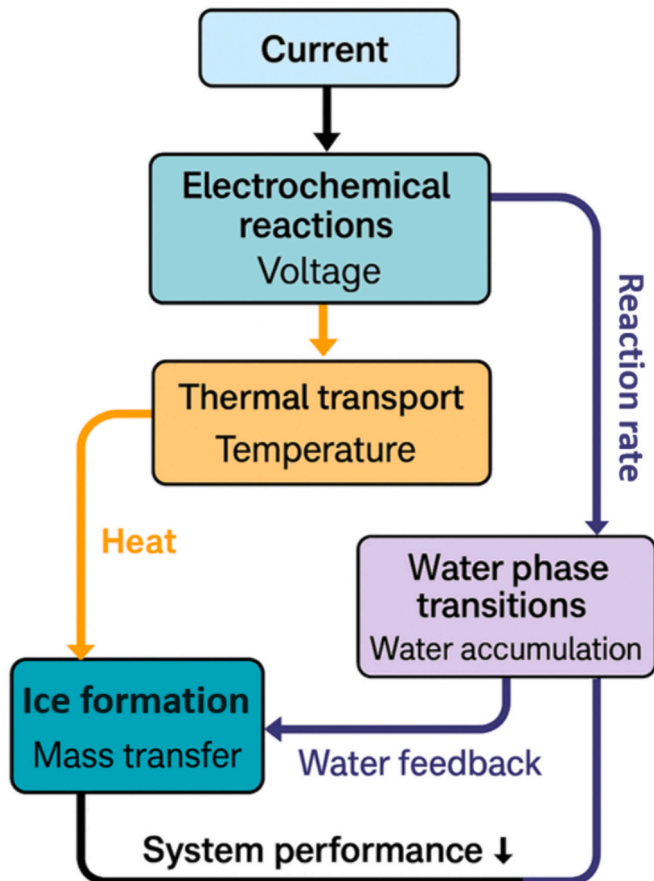


Fig. 10. Coupled multi-physics mechanism.

delays the onset of ice crystallization, and preserves active electrochemical sites across the membrane–electrode interface.

- (3) **Ice formation is governed by coupled thermal and mass transport.** The cathode exhibits more severe icing than the anode, and the CL accumulates significantly more ice than the GDL. Under exponential ramping, the average ice volume fraction in the CCL reaches 0.28, with rapid vertical propagation into the GDL and pronounced ice accumulation near the CCL-CGDL interface. In contrast, the two-stage exponential strategy effectively delays ice onset and restricts its spatial expansion. Intense early loading can lead to ice intrusion into porous structures, while phased or delayed loading strategies can alleviate this effect.
- (4) **Effective cold-start control requires coordinated multi-physics management.** Imbalances between current, heat, and phase change lead to feedback loops that degrade stability. An optimal current ramping strategy should ensure sufficient preheating before aggressive current loading, controlled water production aligned with thermal capacity, and uniform field distribution across both time and space.

In summary, this work introduces a novel current ramping concept, emphasizing that the success of a cold start strategy depends not only on the magnitude of energy input, but more importantly on the temporal and spatial coordination among multiple coupled physical fields. While the present model employs representative kinetic parameters, the underlying principle is broadly applicable. With appropriate tuning of ramping profiles, the proposed strategy can be adapted to different Pt loadings, MEA configurations, and operational scenarios. Although the achieved power output is below typical automotive-rated levels, the strategy demonstrates stable voltage, effective ice suppression, and complete ramping via self-heating alone. These results offer a promising foundation for future scaling toward practical cold start targets through catalyst design, structural optimization, and feedback-based control.

CRediT authorship contribution statement

Guoqiu Liu: Writing – original draft, Methodology, Investigation, Conceptualization. **Lei Xian:** Visualization, Software, Investigation. **Lei Chen:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization. **Wen-Quan Tao:** Writing – review & editing, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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